Science Requirements for

Burning and Suppression of Solids

"BASS"

(Material Flammability in Low-Speed Air Flow at Microgravity Including an Assessment of Inert-Gas Extinguishing Agent: An ISS Glovebox Experiment Utilizing Existing SPICE Hardware with Minor Modifications)

February 2010

Approvals

Paul Ferkul — 3/5/2010

Dr. Paul Ferkul
National Center for Space Exploration Research
Glovebox Investigator

Date

Sandra L. Olson — 3/5/2010

Dr. Sandra Olson
NASA Glenn Research Center
Co-Investigator and Project Scientist

Date

Fumiaki Takahashi — 3/5/2010

Dr. Fumiaki Takahashi
National Center for Space Exploration Research
Co-Investigator

Date

J S T — 3/5/2010

Prof. James T'ien
Case Western Reserve University
Co-Investigator

Date

Robert Hawersaat — 3/5/2010

Robert Hawersaat
NASA Glenn Research Center
Glovebox Program Manager

Date

Background

Flame spread behavior, flammability, and suppression data are of interest to NASA for application to fire safety efforts. The reduced buoyant flows, coupled with low-speed air circulation currents, create an environment quite different than normal Earth gravity. In addition, there are several oxidizer atmospheres proposed for crew-manned vehicles and habitats [NASA CxP-70024], which further complicates the problem. Understanding flammability and suppression in a variety of environments is especially important given NASA's current exploration initiatives.

There is some evidence for the *hypothesis* that materials in microgravity, with adequate ventilation, burn as well if not better than the same material in normal gravity with other conditions being identical (pressure, oxygen concentration, temperature, etc.) To test this hypothesis with any but the thinnest materials requires extended microgravity time – longer than available in drop towers or parabolic aircraft. If this hypothesis is true, it implies serious safety concerns over the utility and margin of safety provided NASA-STD-6001 Test #1.

There are important differences in the suppression of fires in space compared to on Earth. On Earth it is understood that best results are generally obtained when the extinguisher "attacks" the base of the flame, which is both the stabilization point and the point of fresh air entrainment. For a fire burning in microgravity, the best point of application of suppressant may not be immediately apparent, especially for a partially obstructed flame or a wake-stabilized flame. Depending on the geometry of the flame and the characteristics of the extinguisher (distance from flame, dispersion angle) it is possible that the suppressant stream will be ineffective or might actually make the flame worse through the entrainment of oxygen.

Objectives

There are two main *objectives*:

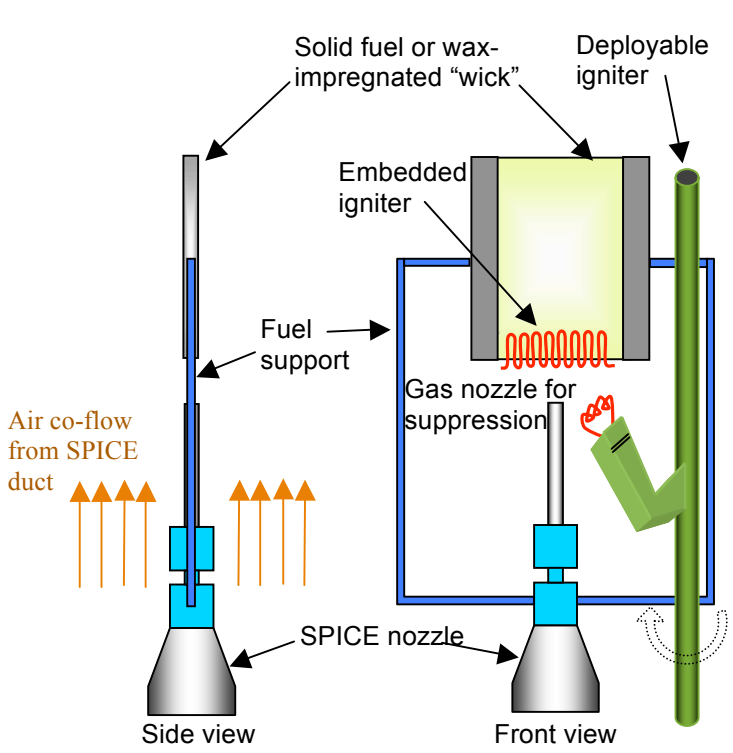
1.) Begin to bridge the gap between the normal gravity NASA-STD-6001 Test#1 method, ground-based microgravity tests, and actual material flammability in space. The relation between these can be investigated by observing flames burning for the longer times available in flight. Practical materials in typical geometries will be examined, including wake flames which have characteristics of a fire burning behind another object (for example, a fire burning in tight area which may not be easily accessible). Fuels can be significantly thicker than the thin sheets studied in ground-based facilities.

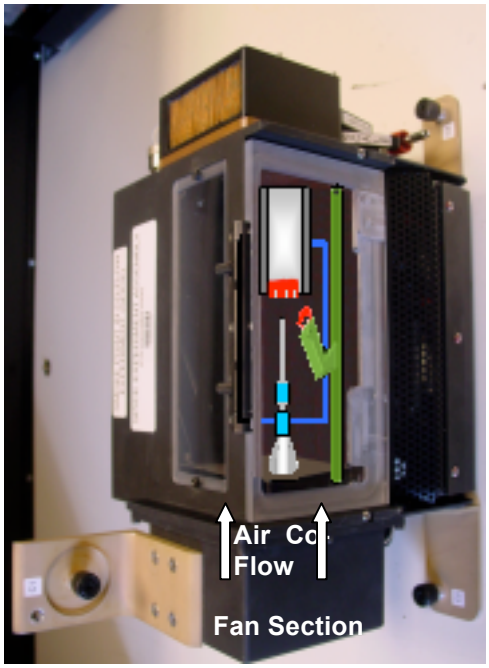
2.) Assess the effectiveness of an inert, gaseous extinguishing agent (similar to that used on ISS) in putting out flames over different materials, geometries, preheating, and flow.

Hardware changes

The plan is to use the existing NASA glovebox SPICE hardware with four changes:
1.) Accommodation of replaceable solid fuel samples
2.) Igniter element adjustable (e.g. by bending wire leads) into the correct position
3.) Nitrogen gas jet for flame extinguishment
4.) Providing a way to shut off the flow completely

Figure 1. Possible Concept:
Fuel Assembly for Flat Sample



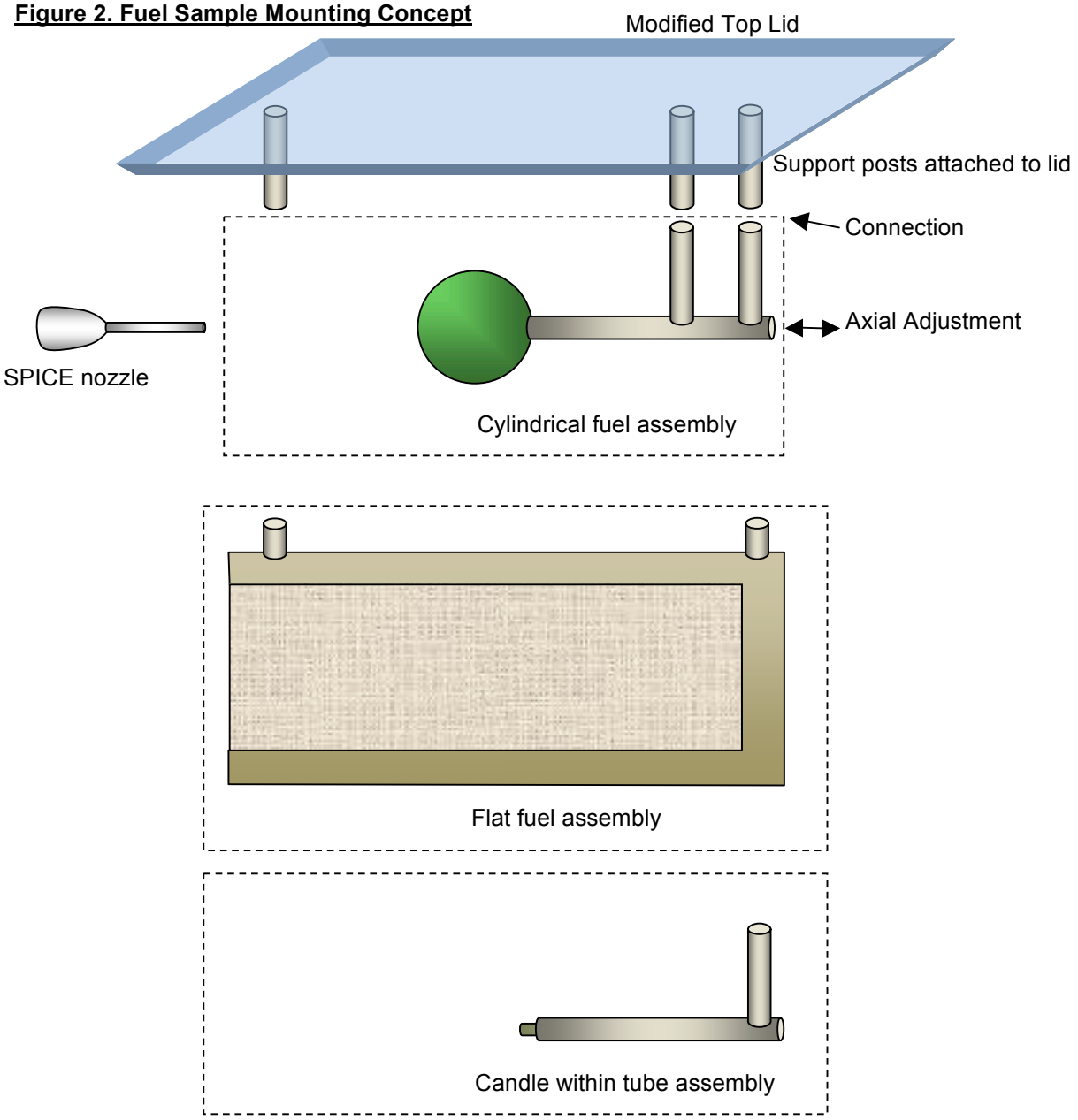


Flat sample shown installed in SPICE hardware

The air co-flow is produced by using a fan to blow air through the duct. An anemometer is used to measure the actual flow speed. The airflow speed can be adjusted from 5 to 50 cm/s. Nitrogen is supplied through the nozzle from pressurized bottles via a mass flow controller, from 0 to 500 cc/min. A radiometer measures flame output. Video and still cameras are used to image the flame.

Multiple fuel assemblies (total of 30, see test matrix) will be used. Figure 1 presents a possible concept for mounting to the existing nozzle structure. Another concept for mounting to the top window is shown in Figure 2.

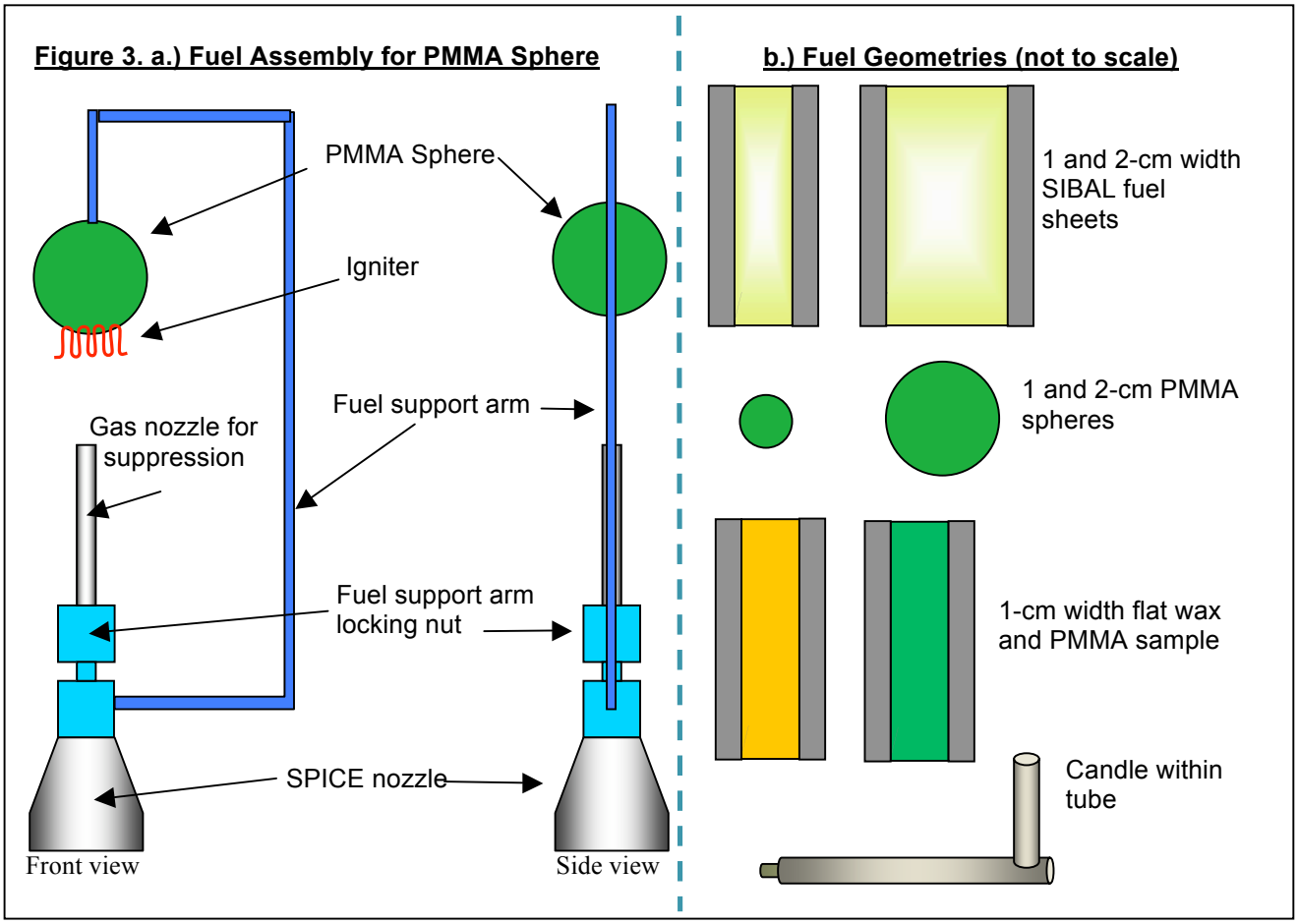
Figure 2. Fuel Sample Mounting Concept



There are three categories of sample: flat sheets, solid spheres, and candles within tubes (Figure 3). For the sheets, concurrent-flow spread rate and limiting flame length are the main interest. The cotton-fiberglass fabric blend SIBAL[1] fuel is our leading candidate for the sample material, since it was specially developed just for this purpose. For some of these spread tests, the suppressant system will be engaged at a gradually

[1] Solid Inflammability Boundary at Low Speed (Principal Investigator: James T'ien) was a previously funded flight project to examine the flammability of a thin solid material burning within an airflow tunnel in microgravity.

increasing level until extinction is reached. A flat slab of fiberglass fabric saturated with wax wax and a one-sided flat PMMA sample are included.



For the solid spheres, flame growth rate, flame shape, and luminosity are the main interests. Again, the suppression system will be engaged after some period of burning. Some of these samples will have inert hemispheres so that a wake flame can be investigated (the inert hemispheres prevent flames from spreading into the forward-stagnation region). Wake flames may occur in real situations where flames are confined behind obstacles and within tight spaces. The burning behavior of these samples and their suppression characteristics are of interest. For the candles within tubes, the effect of suppression amount on the flame structure at different flow speeds will be investigated.

Science Requirements (with reference to SPICE SRD)

Existing SPICE Requirement	BASS Requirement
General requirements	
Configuration: Round gas jet diffusion flame in a co-flowing oxidizer stream	Configuration: Solid fuel materials on individual and exchangeable sample cards
Pressure: Shuttle ambient pressure	---Same---
Ignition: Hot wire	---Same---
Fuel Flow Range: 0 to 500 SCCM N_2 equivalent flow	Nitrogen gas to be used instead of fuel: 0 to 500 SCCM N_2
Oxidizer Velocity: 5 to 50 cm/s	---Same---
Nozzle Diameter: (0.4, 0.8, 1.6 mm)	2 required nozzles: Diameter: (3.2 mm normal length and 3.2 mm short length)
Flame Length: 5 to 100 mm	---Same---
n/a	Flow must be shut off completely as a safety mechanism to extinguish flame
Parameters to be measured: (Accuracy +/- 10% of reading, 4 Hz)	
Fuel Mass Flow: 25 to 500 SCCM	Nitrogen gas to be used instead of fuel: 25 to 500 SCCM N_2
Co-flow Fan Voltage: 0.5 to 12 V	---Same---
Co-flow Velocity: 5 to 50 cm/s	---Same---
Flame Radiation: wavelength visible to near infra-red	---Same---
Imaging Requirements	
Video record of flame: color (FOV: 10 mm of the nozzle to the end of the duct)	Video record of flame: color (FOV: 20 mm upstream of the fuel to the end of the duct)
High resolution color images of flame: 2-3 per test (FOV: 10 mm of the nozzle to the end of the duct; resolution: 3000 by 2000 pixels or greater, 8 bits/pixel in 3 colors or greater). Images stored in raw (uncompressed) mode.	---Same---

Existing SPICE Requirement	BASS Requirement
Acceleration Requirements	
n/a	Testing not to be conducted during thruster fire or spacecraft dock or undock from ISS. Based on the SPICE results, the normal ISS operations provide an adequate acceleration environment.
n/a	When possible, the acceleration environment is to be measured by SAMS during testing.
Ambient Oxygen Concentration Requirements	
n/a	On each day of testing, the oxygen concentration of the ISS cabin atmosphere should be measured. **Testing is restricted to oxygen concentrations at or above 21% O_2.**
n/a	When possible, the oxygen concentration within the MSG working volume shall be known during testing. It should be possible for a crewmember to measure the oxygen concentration within the MSG working volume with a CSA-CP (Compound Specific Analyzer – Combustion Products). **Testing is restricted to oxygen concentrations at or above 21% O_2.**
n/a	The oxygen concentration can be measured by the MCA (Major Constituent Analyzer) or oxygen sensors in the Columbus module or Service Module (i.e., Zarya). The oxygen concentration could also be measured by a crewmember using a CSA-CP (Compound Specific Analyzer – Combustion Products). **If possible, it is desired that the same O_2 measurement sensor be used for all tests to eliminate variation between instruments.**

Existing SPICE Requirement	BASS Requirement
Operation requirements	
Crew sets the co-flow based upon the test matrix and ignites the flame. Once ignited the fuel flow is adjusted until the smoke height is found. High resolution digital flame images are taken above and below the smoke height. The total crew time is estimated to be 20 to 25 hours.	Crew inserts fuel sample and adjusts igniter and fuel base to desired position. The co-flow air is set to the specified value. Once ignited the fuel is allowed to burn for about one minute. High resolution digital flame images are taken. Nitrogen flow is then initiated to extinguish the flame. Total crew time estimate: 30 hours.
n/a	Flow must be shut off completely as a safety mechanism to extinguish flame
Downlink	
Real time video downlink is required for the first few tests and highly desired for subsequent tests. But, it is acceptable for the crew to perform several tests without downlink as long as the video is downlinked before the next series of tests. Video downlink will include fuel flow rate, radiometer readings, and co-flow rate.	Real time video downlink is required for the first few tests and highly desired for subsequent tests. But, it is acceptable for the crew to perform several tests without downlink as long as the video is downlinked before the next series of tests. Video downlink will include nitrogen flow rate, radiometer readings, and co-flow rate.
Glovebox and Mission Resources	
Experiment Power Converter Box, video monitor, late stowage (L-48 hours).	Experiment Power Converter Box, video monitor.

Test Matrix

Sample #	Test ID	Description	Estimated Fuel Burned (mg)	Flow speed (cm/s)	Suppression
1	1	Thin Flat, 2-cm Width SIBAL fuel	216	5	Optional based on results; when flame covers entire sample
2	2		216	10	
3	3		216	20	
4	4		216	50	
5	5	Thin Flat, 1-cm Width SIBAL fuel	108	5	
6	6		108	20	
7	7	Thin Flat, 2-cm Width Nomex III (Blend)	132	30	
8	8		132	50	
9	9	Thin Flat, 2-cm Width WSTF Nomex Natural (HT90-40)	252	30	
10	10		252	50	
11	11	Thin Flat, 2-cm Width Ultem	384	30	
12	12		384	50	
13	13-16	Sphere, 1-cm Diameter PMMA	300	5, 10, 20, 50 (Re-ignition at multiple speeds)	After 1 minute; Extinguishment will be initiated by gradually increasing the mass flow rate of nitrogen in a series of step increments
14	17-20		300		
15	21-24	Hemisphere, 1-cm Diameter (Wake Flame) PMMA	300		
16	25-28		300		
17	29-32	Sphere, 2-cm Diameter PMMA	1200		
18	33-36		1200		
19	37-40	Hemisphere, 2-cm Diameter (Wake Flame) PMMA	1200		
20	41-44		1200		
21	45-48	Thick Flat, 1-cm Width, 2-mm Thick PMMA, One-Sided	119		
22	49-52	Flat fiberglass fabric saturated with wax (paraffin), 1-cm Width 2-mm Thick, One-Sided	100		
23	53	Candle (Japan wax) within tube (front ignition)	200	5 (temporarily 0)	
24	54	Candle (Japan wax) within tube (front ignition)	200	10	
25	55	Candle (Japan wax) within tube (front ignition)	200	30	
26	56	Candle (paraffin) within tube (front ignition)	200	5 (temporarily 0)	
27	57	Candle (Japan wax) within tube (wake ignition)	200	5 (temporarily 0)	
28	58	Candle (Japan wax) within tube (wake ignition)	200	10	
29	59	Candle (Japan wax) within tube (wake ignition)	200	30	
30	60	Candle (paraffin) within tube (wake ignition)	200	5 (temporarily 0)	

Success Criteria

- Minimal Success: 20 total tests including: eight tests with flat samples (four materials at two different flow speeds); four tests with spherical PMMA samples (four different flow speeds with nitrogen extinguishment); four tests with candles (four different flow speeds with nitrogen extinguishment); and four tests of PMMA wake flames (four different flow speeds with nitrogen extinguishment).

- Complete Success: 60 flames as indicated in test matrix.